



# AWS GUARDDUTY



THREAT DETECTION MADE SIMPLE 



**DETECT**  
THREATS  
EARLY



**MONITOR**  
YOUR AWS  
ENVIRONMENT



**SMART**  
THREAT  
INTELLIGENCE



**ALERT**  
& TAKE  
ACTION

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# WHAT IS GUARDDUTY?



Amazon GuardDuty is an intelligent threat detection service that continuously monitors your AWS environment for malicious activity and suspicious behavior.

It uses:



MACHINE  
LEARNING



THREAT  
INTELLIGENCE  
FEEDS



BEHAVIORAL  
ANALYSIS



Think of GuardDuty as a 24/7 security guard watching your AWS account.



GUARDDUTY MONITORS:



CLOUDTRAIL  
LOGS



VPC FLOW  
LOGS



DNS  
LOGS



EKS AUDIT  
LOGS



S3 DATA  
EVENTS



IAM  
ACTIVITY



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# REAL-TIME SCENARIO

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Someone **STOLE** IAM CREDENTIALS



## THE ATTACKER:



Accesses **S3** buckets and steals sensitive data



Scans **AWS** resources to find more targets



Logs in from another country / **unusual** location



Launches **crypto mining** to exploit your resources



Tries to **escalate** privileges and access more services



Without **GuardDuty**, this attack may go unnoticed for **hours** or even **days**!





# GUARDDUTY DETECTS IT!

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GuardDuty continuously analyzes logs and identifies **suspicious** and **malicious** activities.



## WHAT IT ANALYZES



**CloudTrail Logs**  
API calls & user activity



**VPC Flow Logs**  
Network traffic



**DNS Logs**  
DNS queries



**S3 Data Events**  
Object level activity



**EKS Audit Logs**  
Cluster activities



**IAM Activity**  
User & credential usage

## THREATS DETECTED



**Unusual Login Locations**  
Login from new or unfamiliar location



**Malicious IP**  
Communication with known bad IPs



**Port Scanning**  
Scanning ports on your instances



**Crypto Mining**  
Unauthorized crypto mining activity



**Backdoor Activity**  
Indicators of backdoors or persistence



When GuardDuty finds something **suspicious**, it generates a **SECURITY FINDING**.



## FINDING INCLUDES:

- ✓ Severity (Low / Medium / High)
- ✓ Affected Resource
- ✓ Attack Details
- ✓ Recommended Actions





# AUTOMATIC RESPONSE WITH GUARDDUTY

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GuardDuty not only detects threats, it can trigger **automatic actions** to protect your application.



## How It Works?



GuardDuty  
Finds a Threat



EventBridge  
Receives the  
Finding



Lambda  
Runs Automated  
Response



Isolate EC2



Revoke IAM  
Access



Send Alert  
(SNS / Slack)



Protect the  
Application

## Application Protection



GuardDuty monitors the entire  
application environment 24/7.



Users



ALB



EC2 / EKS



Application  
Servers



RDS

### Benefits:

- ✓ Detects threats early
- ✓ No infrastructure to manage
- ✓ Works across multiple AWS services
- ✓ Protects your workloads
- ✓ Integrates with automated incident response



With GuardDuty, you can **detect and respond** to threats  
**before attackers damage** your application.





# KEY TAKEAWAYS

## WHY USE

# GUARDDUTY?

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GuardDuty gives you intelligent threat detection so you can respond early and keep your applications and data safe.

### WHY IT MATTERS?



Detects threats early before damage occurs.



Protects your AWS workloads and data.



24/7 continuous monitoring.



No infrastructure to manage. Fully managed service.



Cost-effective protects your environment continuously.



### IN SHORT:

GuardDuty detects suspicious activities. Automation helps you respond. Your application stays secure.



### COMMON USE CASES

✓ Compromised IAM credentials detection

✓ Unusual login locations

✓ Malicious IPs communicating with your resources

✓ Port scanning & Reconnaissance detection

✓ Crypto mining & unauthorized resource usage



GuardDuty + Your Automation = Stronger Security  
Detect faster. Respond faster. Stay protected.

